



RESEARCH ARTICLE

Impact of integrated nutrient management on nutrient availability, nutrients uptake, growth, yield and quality characteristics of mustard (*Brassica juncea* L.) in light-textured soils

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Abstract

The field study was conducted during the rabi season of 2024–25. The experiment was conducted in a randomized complete block design (RCBD) consisting of 10 treatments and 3 replications, along with a control. The results revealed that integrated nutrient management (INM) significantly enhanced nutrient uptake, as well as the growth, yield and quality characteristics of Indian mustard. The application of 100 % recommended dose of fertiliser (RDF) + farmyard manure (FYM) at 2 t ha⁻¹ + vermicompost at 0.6 t ha⁻¹ resulted in higher plant growth parameters as such plant height (216.03 cm), number of primary branches per plant (8.83), number of secondary branches per plant (18.56) and leaf area index (LAI) (0.97). Moreover, the higher yield and quality parameters of mustard were documented under same treatment, which included higher number of silique per plant (618.70), number of seeds per silique (14.86), test weight (4.61g), harvest index (38 %), seed yield (2569.67 kg ha⁻¹), stover yield per ha (4192.67 kg ha⁻¹), biological yield (6762.33 kg ha⁻¹), oil content (37.95 %), oil yield (975.71 kg ha⁻¹), protein content (19.44 %) and protein yield (499.77 kg ha⁻¹). Furthermore, treatment T₇ enhanced the nutrient availability in seed (nitrogen (N): 3.50, phosphorous (P): 0.62, potassium (K): 0.97 and S: 0.40 kg ha⁻¹) and stover (N: 0.48, P: 0.24, K: 0.41 and S: 0.24 kg ha⁻¹). Similarly, higher nutrient uptake was recorded in seed (N: 90.04, P: 15.94, K: 24.94 and S: 10.29 kg ha⁻¹) and stover (N: 20.15, P: 9.93, K: 17.35 and S: 9.93 kg ha⁻¹) under same treatment. In contrast, minimum values for all these characteristics were observed in the treatment T₁ (Control). The main aim of this investigation was to evaluate the impact of INM practices on nutrients availability, nutrients uptake, growth, yield, quality characteristics of mustard (*Brassica juncea* L.) in light-texture soils. The findings of this investigation revealed that treatment T₇ was found to be the most effective nutrient management practice and to identify suitable INM strategies for sustainable mustard cultivation.

Keywords: macronutrients; oil content; oilseed crops; organic amendments; protein content; sustainable agriculture

Introduction

Indian mustard (*Brassica juncea* L.) is a highly desirable oilseed that is essential for the caloric and mineral nutrition of both humans and animals. Rapeseed and mustard are the second most significant oilseed crops in the country, followed by ground nuts and together account for nearly 33 % of the total oilseed production (1). Indian mustard is mostly cultivated in the states of Uttar Pradesh, Rajasthan, Bihar, Gujarat and Madhya Pradesh. Furthermore, its cultivation is being extended to non-traditional areas of southern states like Tamil Nadu, Karnataka and Andhra Pradesh. In India, mustard is cultivated over an area of 8.06 million hectares with total production of 11.75 million tones and an average productivity of 1458 kg ha⁻¹ (2, 3). India ranks as the fourth largest economy in the world in terms of edible oil production, followed by the United States, China and Brazil. It also imports more oil than any other country, second only to China (1, 4).

Integrated nutrient management (INM) is a sustainable approach that emphasises the responsible and efficient utilisation of all major plant nutrient sources to preserve soil fertility, health and productivity. These consist of crop residues, recyclable wastes, legumes in cropping systems, compost, green manures, animal manures, chemical fertilisers, biofertilisers and other locally accessible nutrient resources. INM is a holistic approach aimed at maintaining soil health and ensuring the optimum availability of plant nutrients for sustained crop productivity by efficiently utilising all possible nutrients sources. Integrated nutrient management plays significant role in enhancing nutrient use efficiency (NUE), increasing food grain production and maintaining soil fertility and improving the farmer's income through integrated and balanced application of fertilisers. Furthermore, INM improves seed yield, improves grain quality and enhance nutrient uptake in mustard through balanced fertilisation utilising the right source, time and method (3, 5, 6). The combined uses of chemical fertilisers with

organic manures further improve soil health and sustainability (7, 8).

In addition, organic manures are indispensable for augmenting crop productivity, soil fertility and crop yields in agriculture. They are environmentally friendly and can be replaced with 25 % chemical fertilisers that are non-toxic to human health and enable farmers to generate the highest possible crop yields. The application of organic manure in conjunction with chemical fertilisers result in a substantial increase in the mineral content of the soil, the water-holding capacity of the field and the uptake of nutrients.

Additionally, the crop yield is increased by several other factors, including root development, vegetative growth and nitrogen (N) fixation, as reported by (9, 10). The growth of rapeseed crops is facilitated by phosphorus (P) fertilisation, whereas its deficiency inhibits the growth of the roots and aerial portions. Potash is responsible for activating a diverse array of enzyme systems in rapeseed mustard. Sulphur is appropriately named the fourth major element of the plant, as it is a component of three amino acids and contributes to the synthesis of chlorophyll and oils. The use of secondary major nutrients and micronutrients resulted in a significant response of oilseed, with a residual effect in the cropping system (6, 11). Low fertility and nutrient loss are common in light-textured soils, which limits the amount of mustard that can be grown.

The use of chemical fertilisers alone have a negative effect on soil health and little is known about how INM functions in mustard in these kinds of soils. The gap is addressed in this study by investigating INM as a sustainable method for enhancing crop productivity and soil fertility. This investigation aims to assess the influence of INM on the availability of soil nutrients, the absorption of nutrients, the growth and the yield of mustard (*Brassica juncea* L.) in light-textured soils. Furthermore, the study aims to determine sustainable management techniques that can enhance mustard productivity and seed quality, as well as the effects of different nutrient sources on soil fertility and crop performance.

Materials and Methods

Experimental location

A field experiment was conducted at the Experimental Research Farm, School of Agriculture, Sanjeev Agrawal Global Educational University, Bhopal (Madhya Pradesh), during the rabi season of 2024–25. Geographically, the experimental site is located at an elevation of 500 m above mean sea level, with a longitude of 77.24° E and an altitude of 35.75° N. The soil of the experimental site was black soil. The soil had a silty clay loam texture, its pH was 7.5, its electrical conductivity (EC) was 0.32 dS/m, its organic carbon (OC) content was 0.049 % and its available nutrient status was N: 221.9 kg ha⁻¹, phosphorus (P): 16.7 kg ha⁻¹, Potassium (K): 248.5 kg ha⁻¹ and sulphur (S): 13.8 kg ha⁻¹. Rainfall observed throughout the winter season was 11.40 mm.

Experimental details

The experiment was set up in a randomized complete block design (RCBD) with 10 INM treatment combinations, one of which served as a control. Each treatment was replicated three times. For the experiment, the recommended amounts of N, P, K and S fertilisers 120, 50, 40 and 30 kg ha⁻¹ each were used evenly. Potassium ureate, urea and single superphosphate were used as fertiliser sources.

Nitrogen was applied in two separate applications: half as basal and the remaining half in two equal doses, the first at 30 days after sowing and the second at flowering. Phosphorus and potash were administered as basal doses. The farm yard manure (FYM) and vermicompost were mixed into the soil according to the treatments before it was planted.

Additionally, the soil was treated with biofertilisers such as *Azotobacter* in accordance with the treatment specifications prior to sowing.

Treatment notations	Treatment combinations
T ₁	Control
T ₂	100 % RDF
T ₃	100 % RDF + S at 30 kg ha ⁻¹
T ₄	125 % RDF + S at 30 kg ha ⁻¹
T ₅	100 % RDF + FYM at 4 t ha ⁻¹
T ₆	100 % RDF + vermicompost at 1.25 t ha ⁻¹
T ₇	100 % RDF + FYM at 2 t ha ⁻¹ + vermicompost at 0.6 t ha ⁻¹
T ₈	75 % RDF + FYM at 4 t ha ⁻¹
T ₉	75 % RDF + vermicompost at 1.25 t ha ⁻¹
T ₁₀	75 % RDF + FYM at 2 t ha ⁻¹ + vermicompost at 0.6 t ha ⁻¹

RDF: recommended dose of fertilisers; FYM: farmyard manure.

Treatment details

The following observations were recorded during experimentation.

Plant growth parameters

Plant height: Plant height (cm) was measured from the ground level to the apical tip of the main stem in five randomly selected plants at 30-day intervals, from 30 to 120 days after sowing and mean value for each treatment was expressed in centimetres.

Numbers of primary and secondary branches per plant: The primary and secondary branches were counted on five randomly selected plants each plot and mean number of branches per plant was recorded.

Leaf area (cm²): The leaf area index (LAI) of mustard plant was determined by collecting five leaves from five randomly marked plants in each treatment and measuring their leaf area using the following formula.

$$\text{Leaf area index (LAI)} = \frac{\text{Leaf area}}{\text{Ground area}} \quad (\text{Eqn. 1})$$

Yield parameters

Studies were done on five randomly chosen plants from each net plot at the start of the experiment after they had been harvested. Before harvesting the net plots, these marked plants were cut and removed from the field for further analysis.

Number of siliquae per plant: the number of siliquae on five marked plants was counted and their mean value was expressed as siliqua per plant.

Number of seeds per siliqua: In the same way, the mean count of five siliqua from each marked plant was taken and the average values were expressed as seeds per siliqua.

Test weight (g): Using a digital balance, a random sample of 1000 seeds from each net plot were weighed and recorded in grams. The values were expressed as test weight.

Harvest index (%): The harvest index was calculated using the formula as given below (12):

$$\text{Harvest index} = \frac{\text{Economic yield (kg/ha)}}{\text{Biological yield (kg/ha)}} \quad (\text{Eqn. 2})$$

Seed yield (kg ha⁻¹): The seeds from every net plot were threshed and cleaned. The yield was recorded in kilograms per plot and subsequently converted to kilograms per hectare using the enclosed formula.

Seed yield (kg/ha) =

$$\frac{\text{Yield obtained from net plot (kg)}}{\text{Net plot area}} \times 10000 \quad (\text{Eqn. 3})$$

Stover yield (kg ha⁻¹): The stover yield was calculated by subtracting the seed yield from the total biomass produced (seed + stover) in each plot. The stover yield in kilograms from each net plot was subsequently converted to quintals per hectare using the standard conversion factor 1000.

Biological yield (kg ha⁻¹): The biological yield was determined by subtracting the seed yield from the total biomass produced (seed + stover) in each plot. Afterward, the standard conversion factor 1000 was employed to convert the biological yield in kilograms from each net plot to quintals per hectare.

Seed quality parameters

The oil content (%): Initially, the seed samples were oven-dried at 70 °C to eliminate moisture. Afterward, the seeds were crushed in pestle and mortar and the oil content was determined using the Soxhlet apparatus method (13).

Oil content =

$$\frac{\text{Weight of flask + petroleum ether extract - Weight of flask with oil}}{\text{Weight of sample (g)}} \times 100 \quad (\text{Eqn. 4})$$

Oil yield (kg ha⁻¹): It was calculated by multiplying the oil content by the mustard seed yield using the formula below.

$$\text{Oil yield (kg/ha)} = \frac{\text{Oil content (\%)} \times \text{seed yield (kg ha}^{-1}\text{)}}{100} \quad (\text{Eqn. 5})$$

Protein content (%): The protein content of mustard seed was estimated using a near infrared analyser and recorded for each treatment separately.

Protein yield (kg ha⁻¹): The protein yield (kg ha⁻¹) was calculated by multiplying the seed yield (kg ha⁻¹) by the corresponding protein content (%) of mustard seed and dividing the result by 100. The following formula was used to analyse protein content in mustard seed and the results were expressed as protein yield (kg ha⁻¹)

Protein yield (kg/ha) =

$$\frac{\text{Seed yield (kg/ha)} \times \text{protein content (\%)}}{100} \quad (\text{Eqn. 6})$$

Nitrogen (N), phosphorus (P), potassium (K) and sulphur (S) content in seed and stover and their uptake by seed and stover of mustard

The total uptake of N, P, K and S concentration in seed and stover were estimated using standard procedures (14). To assess the impact of INM on yield, quality and the nutrient composition of mustard, plant samples collected at harvest were air-dried and afterward ground using a grinder fitted with stainless steel blades to avoid micronutrient contamination. The concentrations of N, P, K and S in both seed and stover samples of mustard were then estimated.

The gathered samples were evaluated for available N

(kg ha⁻¹) using the standard procedure through the alkaline potassium permanganate method and the results were expressed in (kg ha⁻¹) (15). Available P (kg ha⁻¹) in the collected samples was analysed using the standard Olsen's method and the results were expressed in (kg ha⁻¹) (16). Available K (kg ha⁻¹) in the soil samples were determined using the flame photometric technique. Sulphur content was estimated through the turbidimetric method. Plant samples were digested using a tri-acid mixture composed of nitric, perchloric and hydrochloric acids. The addition of barium chloride facilitated the development of turbidity, which was then measured using a colorimeter. The sulphur concentration was expressed on a dry weight basis as a percentage.

The formula given below is used to determine the uptake of Nitrogen, phosphorus, potassium and sulfur nutrients by grain and straw:

Uptake of N, P, K and S (kg/ha) =

$$\frac{\text{Yield seed or Stover (q/ha)} \times \text{Nutrient conc. (\%)} \text{ seed or stover}}{100} \quad (\text{Eqn. 7})$$

Statistical analysis

The OPSTAT software (version 6.8) was used to do the statistical analysis of the data collected at different stages of growth for this study. The significance of the results was determined at a threshold of $p \leq 0.05$. The critical difference was employed to compare the treatment means.

Results and Discussion

Plant growth characteristics

The data on the plant growth characteristics of mustard crops, including the number of primary branches per plant, the number of secondary branches per plant and the LAI, were recorded during the investigation period (Table 1). During all growth stages, these characteristics demonstrated substantial variation in response to varying nutrient management practices. The findings demonstrate that the growth characteristics of mustard plants were significantly enhanced by the combined application of 100 % recommended dose of fertilisers (RDF), FYM at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹. Among the various treatment, the treatment T₇ (100 % RDF + FYM at 2 t ha⁻¹ + vermicompost at 0.6 t ha⁻¹) were recorded maximum values for plant height (216.03 cm), number of primary branches per plant (8.83), number of secondary branches per plant (18.56) and LAI (0.97). The minimum values of these characteristics were recorded under treatment T₁ (control). The enhancement in growth characteristics may be attributed to a more favourable soil environment that incorporates a nutrient management system. The advantages may have been attributed to the combined application of inorganic and organic manure, which not only met the immediate nutrient needs but also created an optimal soil environment for the cultivation of plants. Significant enhancements in plant growth have been achieved as a result of favourable soil conditions, which are primarily due to the presence of N. Nitrogen is very important for making chlorophyll, dividing cells and making cells longer. Its role in making DNA and proteins makes sure that cells can replicate and stay structured and its contribution to chlorophyll makes photosynthesis better. The combined effect of these factors results in enhanced growth parameters, including increased biomass, expanded leaf area and

Table 1. Influence of integrated nutrient management on plant growth characteristics of mustard

Treatment notation	Plant height (cm)	Number of primary branches/plants	Number of secondary branches/plants	Leaf area index
T ₁	154.93	5.83	13.45	0.74
T ₂	175.73	6.53	14.55	0.78
T ₃	176.93	6.73	14.95	0.80
T ₄	181.73	7.03	15.55	0.85
T ₅	195.97	7.73	17.26	0.87
T ₆	196.63	8.23	17.66	0.90
T ₇	216.03	8.83	18.56	0.97
T ₈	195.13	7.37	16.06	0.91
T ₉	199.13	7.53	16.76	0.87
T ₁₀	204.07	8.53	18.16	0.88
S. Em. ±	1.18	0.11	0.07	0.03
C.D. ($p = 0.05$)	3.54	0.32	0.21	0.08

overall improved plant vigour. The current experiment's results are almost in agreement with those of (5, 17–20).

Yield characteristics

The data on mustard yield characteristics, which include the number of siliqua/plant, number of seeds/silique, test weight (g), harvest index, seed yield (kg ha^{-1}), stover yield (kg ha^{-1}) and biological yield (kg ha^{-1}), were documented during the study period (Table 2). During all growth stages, these characteristics exhibited substantial variation in response to varying nutrient management practices. The findings demonstrated that the mustard yield characteristics were significantly enhanced by the combined application of 100 % RDF, FYM at 2 t ha^{-1} and vermicompost at 0.6 t ha^{-1} . Among the various treatment, the treatment T₇ (100 % RDF + FYM at 2 t ha^{-1} + vermicompost at 0.6 t ha^{-1}) recorded maximum values for number of siliqua per plant (618.70), number of seeds per siliqua (14.86), test weight (4.61g), HI (38 %), seed yield ($2569.67 \text{ kg ha}^{-1}$), stover yield ($4192.67 \text{ kg ha}^{-1}$) and biological yield ($6762.33 \text{ kg ha}^{-1}$). In contrast, minimum values for these characteristics were obtained under the treatment T₁ (control). This substantial improvement in yield characteristics, such as the number of siliquae per plant, the number of seeds per siliqua, the test weight and the HI, may be attributed to the combined application of inorganic and organic fertilisers. The integration of inorganic fertilisers with organic manures improves the physical, chemical and biological properties of the soil, thereby facilitating the optimal growth of photosynthetic organs, efficient nutrient translocation and enhanced photosynthetic activity. This ultimately leads to a larger plant frame that can support more yield-contributing traits. These findings are in agreement with the earlier findings (5, 21–25).

Seed yield, stover yield and biological yield: The enhanced seed yield, stover yield and biological yield may be attributed to the

combined application of inorganic and organic fertilisers, which leads to an increase in root growth. This resilient root system enables the soil to absorb moisture and nutrients at an optimal rate, thereby enhancing dry matter production. The resulting translocation of photosynthates from leaves to the seed-bearing siliquae further ensures the development of high-quality seeds. It is probable that the seed size, seed weight, number of seeds has increased because of the synthesis process, additional carbohydrates and the integrated nutrition supply. Increasing seed yield has been made possible by the synergistic relationship between organic and inorganic fertilizers. These results agree with earlier findings of (5, 21–26). Additionally, using 75 % chemical fertiliser and 25 % organic manure together led to much higher yields of mustard seeds, stover and living things compared to the other treatments (24). An increase in all yield attributes was the result of the increased availability of photosynthesis and nutrients to the plant's reproductive structures. This increase could be used to improve the final yield of the mustard crop. Furthermore, it was claimed that the sink capacity of a plant is primarily determined by vegetative growth, which is positively impacted by the application of N fertilisers and the supply of photosynthesis, which is essential for the formation of plant yield components. These results are consistent with the previous findings (19).

Quality characteristics

The data on the quality characteristics of mustard, as well as oil content, oil yield (kg ha^{-1}) and protein content (kg ha^{-1}), were calculated during the investigation period and are illustrated in Table 3. These characteristics revealed substantial variation under different nutrient management practices at all growth stages. The outcomes indicated that the combined application of 100 % RDF along with FYM at 2 t ha^{-1} and vermicompost at 0.6 t ha^{-1} resulted in a significant enhancement in mustard quality characteristics. Among the various treatment, the treatment T₇ (100 % RDF + FYM at 2 t ha^{-1} + vermicompost at 0.6 t ha^{-1}) recorded maximum values for oil content (37.95 %), oil yield ($975.71 \text{ kg ha}^{-1}$), protein content (19.44 %) and protein yield ($499.77 \text{ kg ha}^{-1}$). In contrast, minimum values for these characteristics were estimated under the treatment T₁ (control). The increase in the quality characteristics of mustard may be attributed to the application of 100 % RDF in conjunction with FYM at 2 t ha^{-1} and vermicompost at 0.6 t ha^{-1} . Farmyard manure (FYM) and vermicompost are abundant sources of both macronutrients and micronutrients, which ultimately contribute to improved seed yield and quality. The implementation of micronutrients in these organic amendments guarantees a gradual and consistent release of nutrients, thereby facilitating the physiological functions of the plant. It also increases the oil content in seeds, resulting in a general increase in the total oil yield. Similar

Table 2. Influence of integrated nutrient management on yield characteristics of mustard

Treatment notation	Number of siliqua/plant	Number of seeds/silique	Test weight (g)	Harvest index (%)	Seed yield (kg ha^{-1})	Stover yield (kg ha^{-1})	Biological yield (kg ha^{-1})
T ₁	446.60	11.84	3.88	35.65	1893.00	3417.00	5310.00
T ₂	527.73	12.55	4.19	35.76	2028.67	3645.00	5673.67
T ₃	568.60	13.75	4.28	36.39	2148.00	3755.33	5903.33
T ₄	578.73	12.69	4.21	37.08	2267.33	3847.33	6114.67
T ₅	588.64	12.75	4.31	37.79	2349.33	3866.67	6216.00
T ₆	598.30	13.25	4.28	36.22	2258.33	3976.00	6234.33
T ₇	618.70	14.86	4.61	38.00	2569.67	4192.67	6762.33
T ₈	591.13	12.40	4.18	37.73	2343.67	3868.00	6211.67
T ₉	576.46	12.56	4.18	37.16	2276.00	3848.33	6124.33
T ₁₀	585.07	12.57	4.27	36.71	2243.33	3867.67	6111.00
S. Em. ±	0.72	0.03	0.02	0.10	15.63	21.38	34.35
C.D. ($p = 0.05$)	2.17	0.09	0.07	0.30	46.80	64.0	102.84

results agree with the earlier findings (27–29). However, oil yield in crops is influenced not only by seed yield but also by the biochemical processes governing oil synthesis, which are highly sensitive to nutrient balance and environmental stress. A sufficient and balanced supply of macronutrients creates favourable conditions for seed development, while micronutrients play a crucial role in improving oil yield. Among these, sulfur plays a direct role in the production of enzymes and amino acids needed for the metabolism of fatty acids and the biosynthesis of oil. Thus, sulfur significantly contributes to increased oil content and overall oil yield. These results are supported by the reports of earlier studies (30, 31). Therefore, the synergistic effect of increased seed yield and higher oil content may be responsible for the increase in mustard oil yield. Similar findings have also been reported (3, 17). Moreover, the application of 100 % RDF in conjunction with FYM at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹ may have increased the availability of N in the soil, resulting in a substantial increase in protein content. This

Table 3. Influence of integrated nutrient management on quality characteristics of mustard

Treatment notation	Oil content (%)	Oil yield (kg ha ⁻¹)	Protein content (%)	Protein yield (kg ha ⁻¹)
T ₁	33.42	632.64	17.68	334.62
T ₂	35.93	728.85	18.34	372.14
T ₃	36.55	785.10	18.42	395.60
T ₄	36.69	831.97	18.34	415.91
T ₅	36.69	861.98	18.45	433.54
T ₆	37.33	843.07	18.78	424.06
T ₇	37.95	975.71	19.44	499.77
T ₈	36.32	851.31	18.68	437.74
T ₉	36.35	827.33	18.78	427.36
T ₁₀	35.99	807.46	18.91	424.23
S. Em. ±	0.40	12.16	0.23	6.76
C.D. (<i>p</i> = 0.05)	1.19	36.42	0.68	20.24

is due to the fact that N is the primary component of amino acids, which are the foundation of protein. Similar findings are in agreement with the earlier findings (32, 33).

Nutrient concentration in seed and stover

The nutrient contents (N, P, K and S) in mustard seed and stover were evaluated after harvest and the results are summarized in Table 4.

The application of organic and inorganic fertilizers, as well as the fertility levels of the soil, considerably affected the concentration of N, P, K and S in Indian mustard. The concentration of these nutrients was significantly enhanced by the combined application of 100 % RDF, FYM at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹. The maximum values of available N, P, K and S in seed (3.50, 0.62, 0.97 and 0.40 kg ha⁻¹ respectively) and in stover (0.48, 0.24, 0.41

and 0.24 kg ha⁻¹, respectively) were documented under treatment T₇ (100 % RDF + FYM at 2 t ha⁻¹ + vermicompost at 0.6 t ha⁻¹). In contrast, the minimum nutrient availability was observed in treatment T₁ (control), which showed a significant reduction compared with the rest of the treatments. The essential macronutrients (N, P, K and S) improved nutrient availability by increasing their concentration in the seed and stover of mustard, thereby making them more available even within a limited active root zone. Moreover, the application of 100 % RDF along with FYM and vermicompost improved nutrient use efficiency. FYM and vermicompost contribute organic acids that aid in the chelation and availability of macronutrients in the seed and stover of mustard. They also support microbial activity, which facilitates the decomposition of organic matter and the subsequent release of essential nutrients such as N, P, K and S. Additionally, the direct addition of N, P, K and micronutrients to the soil enriched the nutrient pool and further improved nutrient availability. These results agree with the findings of previous studies (34, 35).

Nutrient uptake

The data on nutrient uptake by mustard seed and stover, as well as N, P, K and S, were calculated following the crop's harvest and are presented in Table 5. The fertility of the soil, the availability of macro and micronutrients in both seed and stover and the application of organic and inorganic fertilisers all significantly influenced the uptake of N, P, K and S in Indian mustard.

The nutrient uptake by mustard seed and stover was significantly enhanced by the application of 100 % RDF, FYM at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹. The maximum nutrient uptake such as N, P, K and S in seed (90.04, 15.94, 24.94 and 10.29 kg ha⁻¹, respectively) and in stover (20.15, 9.93, 17.35 and 9.93 kg ha⁻¹, respectively) was recorded under treatment T₇ (100 % RDF + FYM at 2 t ha⁻¹ + vermicompost at 0.6 t ha⁻¹). In contrast, the lowest nutrient uptake was observed in treatment T₁ (control), which showed a significant reduction compared with the rest of the treatments. The utilisation of macronutrients (N, P, K and S) was enhanced by the increased availability of nutrients in the soil as a result of the application of 100 % RDF, FYM at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹. This effect was observed even within a comparatively smaller active root zone. Furthermore, the utilisation of 100 % RDF in conjunction with FYM and vermicompost improved the efficiency of nutrient uptake. The organic acids provided by FYM and vermicompost aid in the chelation and accessibility of micronutrients such as boron and zinc. Additionally, it promotes microbial activity, which aids in the digestion of organic matter and the release of nutrients. Furthermore, FYM and vermicompost enhance the water-holding capacity and soil texture, thereby fostering a more conducive environment for nutrient absorption

Table 4. Influence of integrated nutrient management on nutrient availability in seed and stover of mustard

Treatment notation	Concentration of nutrients in seed (kg ha ⁻¹)				Concentration of nutrients in stover (kg ha ⁻¹)			
	N	P	K	S	N	P	K	S
T ₁	2.99	0.53	0.71	0.34	0.39	0.16	0.27	0.19
T ₂	3.08	0.57	0.86	0.36	0.43	0.18	0.32	0.20
T ₃	3.11	0.55	0.84	0.38	0.45	0.19	0.35	0.21
T ₄	3.16	0.57	0.87	0.38	0.44	0.20	0.37	0.22
T ₅	3.17	0.58	0.82	0.36	0.45	0.21	0.38	0.20
T ₆	3.18	0.59	0.84	0.35	0.45	0.22	0.37	0.21
T ₇	3.50	0.62	0.97	0.40	0.48	0.24	0.41	0.24
T ₈	3.19	0.56	0.85	0.36	0.45	0.21	0.37	0.21
T ₉	3.20	0.57	0.90	0.37	0.43	0.22	0.38	0.22
T ₁₀	3.22	0.55	0.91	0.37	0.45	0.22	0.39	0.21
S. Em. ±	0.03	0.01	0.03	0.01	0.01	0.01	0.01	0.01
C.D. (<i>p</i> = 0.05)	0.09	0.03	0.08	0.03	0.03	0.02	0.03	0.30

Table 5. Influence of integrated nutrient management on nutrient uptake in seed and stover of mustard

Treatment notation	Uptake by seed (kg ha ⁻¹)				Uptake by straw (kg ha ⁻¹)			
	N	P	K	S	N	P	K	S
T ₁	56.66	9.97	13.38	6.37	13.44	5.47	9.23	6.38
T ₂	62.55	11.63	17.45	7.30	15.55	6.68	11.66	7.41
T ₃	66.73	11.74	17.97	8.23	16.90	7.26	13.14	7.89
T ₄	71.65	12.85	19.65	8.54	16.80	7.57	14.36	8.34
T ₅	74.40	13.70	19.26	8.54	17.27	7.99	14.56	7.86
T ₆	71.89	13.40	19.05	7.90	18.02	8.61	14.84	8.35
T ₇	90.04	15.94	24.94	10.29	20.15	9.93	17.35	9.93
T ₈	74.69	13.13	19.84	8.52	17.28	8.12	14.31	8.25
T ₉	72.76	12.90	20.48	8.35	16.68	8.47	14.62	8.47
T ₁₀	72.16	12.26	20.49	8.23	17.28	8.38	14.96	7.99
S. Em. ±	1.03	0.26	0.57	0.57	0.41	0.30	0.40	0.41
C.D. (p = 0.05)	3.03	0.79	1.69	1.69	1.21	0.91	1.20	1.21

and root growth. These combined effects resulted in enhanced nutrient availability and uptake, which in turn facilitated improved plant metabolism, higher yield and a higher quality of seed and oil (35, 36).

Conclusion

Based on the experiment, it is concluded that the combined application of 100 % recommended dose of fertilisers (RDF) along with farmyard manure (FYM) at 2 t ha⁻¹ and vermicompost at 0.6 t ha⁻¹ significantly improved nutrient availability and uptake, which proved highly beneficial for improving yield, quality attributes and overall productivity. The maximum uptake of nitrogen (N), phosphorous (P), potassium (K) and sulphur (S) were also observed in same treatment. Thus, it is advised to implement integrated nutrient management INM, which integrates both organic and inorganic sources, to optimise the quality and yield of mustard cultivation. The utilisation of fertilisers is optimised and productivity is enhanced through balance nutrient application, suitable for commercial mustard cultivation.

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Authors' contributions

YK, YS, GG and AT conducted the experiments. YK, YS, GG, AT, NP, UV, ABS and DK contributed significantly to the design of the study. The manuscript was written by all authors. All authors have read and approved the final Manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interests to declare.

Ethical issues: None

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